Definition 2.3.5. A representation $V \neq 0$ of A is **irreducible** (or **simple**) if the only subrepresentations of V are 0 and V.

Definition 2.3.6. Let V_1, V_2 be two representations of an algebra A. A **homomorphism** (or **intertwining operator**) $\phi : V_1 \to V_2$ is a linear operator which commutes with the action of A, i.e., $\phi(av) = a\phi(v)$ for any $v \in V_1$. A homomorphism ϕ is said to be an **isomorphism of representations** if it is an isomorphism of vector spaces. The set (space) of all homomorphisms of representations $V_1 \to V_2$ is denoted by $\mathrm{Hom}_A(V_1, V_2)$.

Note that if a linear operator $\phi : V_1 \to V_2$ is an isomorphism of representations, then so is the linear operator $\phi^{-1} : V_2 \to V_1$ (check it!).

Two representations between which there exists an isomorphism are said to be isomorphic. For practical purposes, two isomorphic representations may be regarded as "the same", although there could be subtleties related to the fact that an isomorphism between two representations, when it exists, is not unique.

Definition 2.3.7. Let V_1, V_2 be representations of an algebra A. Then the space $V_1 \oplus V_2$ has an obvious structure of a representation of A, given by $a(v_1 \oplus v_2) = av_1 \oplus av_2$. This representation is called the **direct sum** of V_1 and V_2.

Definition 2.3.8. A nonzero representation V of an algebra A is said to be **indecomposable** if it is not isomorphic to a direct sum of two nonzero representations.

It is obvious that an irreducible representation is indecomposable. On the other hand, we will see below that the converse statement is false in general.

One of the main problems of representation theory is to classify irreducible and indecomposable representations of a given algebra up to isomorphism. This problem is usually hard and often can be solved only partially (say, for finite dimensional representations). Below we will see a number of examples in which this problem is partially or fully solved for specific algebras.